CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 21: Trigonometrical Identities

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Prove that $(1 - \sin^2 A)\sec^2 A = 1$.

2. Prove that $\tan^2 A - \sin^2 A = \tan^2 A \sin^2 A$.

3. Prove that $\dfrac{1}{1+\sin A} + \dfrac{1}{1-\sin A} = 2\sec^2 A$.

4. Prove that $\dfrac{\sin A}{1+\cos A} + \dfrac{1+\cos A}{\sin A} = 2\,\mathrm{cosec}\, A$.

5. Prove that $(\sin A + \cos A)^2 + (\sin A - \cos A)^2 = 2$.

6. Prove that $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A$.

7. Evaluate $\dfrac{\cos 38^\circ}{\sin 52^\circ} - \dfrac{\tan 19^\circ}{\cot 71^\circ}$.

8. Evaluate $\sin^2 34^\circ + \sin^2 56^\circ + 2\tan 18^\circ \tan 72^\circ$.

9. If $\tan 2A = \cot(A - 18^\circ)$, where $2A$ is acute, find A.

10. If $A + B = 90^\circ$, prove that $\sqrt{\dfrac{\tan A \tan B + \tan A \cot B}{\sin A \sec B}} = \sec A$.

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Answer Key

1. $\cos^2 A \times \dfrac{1}{\cos^2 A} = 1$
2. LHS $= \sin^2 A \left(\dfrac{1}{\cos^2 A} - 1\right) = \sin^2 A \times \dfrac{\sin^2 A}{\cos^2 A}$
3. LHS $= \dfrac{2}{1 - \sin^2 A} = \dfrac{2}{\cos^2 A}$
4. Numerator $= \sin^2 A + 1 + 2\cos A + \cos^2 A = 2(1 + \cos A)$; LHS $= \dfrac{2}{\sin A}$
5. $1 + 2\sin A \cos A + 1 - 2\sin A \cos A = 2$
6. LHS $= \sec^2 A(\sec^2 A - 1) = (1 + \tan^2 A)\tan^2 A$
7. $1 - 1 = 0$
8. $1 + 2 = 3$
9. $90^\circ - 2A = A - 18^\circ$, so $A = 36^\circ$
10. $\tan B = \cot A$, $\cot B = \tan A$, $\sec B = \operatorname{cosec} A$: inside the root $= \dfrac{1 + \tan^2 A}{1} = \sec^2 A$

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